James Akl

Engineer & Researcher

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SUMMARY

I bring a strong background in **Engineering** and **Mathematics**, along with a special focus in **Robotics**, **AI**, **Control**, and **Data**.
In my work, I am **structured**, **meticulous**, and **rigorous** — I am routinely invested in raising and maintaining **high standards**.
I am excellent at **analyzing problems** and requirements, as well as devising **creative solutions** to address them.
I excel at **project leadership**, planning, and management in technical and R&D settings — I enjoy leading and collaborating.
My technical skills cover **software development** (C++, Python), **mathematical modeling**, **system design**, and **prototyping**.
I seek opportunities in **technology**, **business**, and **industry** to create value via autonomous and intelligent systems.

EDUCATION Coursework

PhD — **Robotics Engineering** — Worcester Polytechnic Institute — GPA 4.00/4.00 — **Aug 2019** – **Aug 2023**
Dissertation: "Robot Autonomy for Scrap Cutting in Metal Recycling"

MS — **Robotics Engineering** — Worcester Polytechnic Institute — GPA 4.00/4.00 — **Aug 2019** – **May 2021**
Focus: Robot dynamics, Robot control, Artificial intelligence, Data science, Applied mathematics

BEng — **Mechanical Engineering** — Lebanese American University — GPA 3.72/4.00 — **Sep 2015** – **May 2019**
Thesis: "Robotic Manipulator for Virtual Reality Haptic Feedback"

Minor — **Mathematics** — Lebanese American University — GPA 4.00/4.00 — **Sep 2015** – **May 2018**
Focus: Analysis, Optimization, Linear algebra, Numerical analysis, Probability & Statistics, Modeling

EXPERIENCE

Applied Scientist Co-op — Amazon Robotics
Innovation Lab (BOS27) — **Aug 2023** – **PRESENT**

Conducting robotics research to design and implement cutting-edge algorithms for solving challenging problems.

Doctoral Researcher — Worcester Polytechnic Institute
Manipulation & Environmental Robotics Lab — **Aug 2019** – **Aug 2023**

Designing and implementing **innovative solutions** and **novel algorithms** to solve challenging technical problems.
Leading and **managing research projects**, supervising MS students, and collaborating in multidisciplinary teams.
Developing **software** for prototypes, simulations, and experiments (C++, Python, ROS, Bash, Linux, Git, PCL, OpenCV, …).
Engineering **autonomous systems** for industrial applications using **robot vision**, **control**, **planning**, and **machine learning**.
My work resulted in **8 research publications** in top-tier conferences/journals, as well as **3 patents** filings.

Research Mentor — Worcester Polytechnic Institute
Robotics Engineering Department / NSF REM Program — **May 2022** – **Jul 2022**

Mentored students in **research methodology**, **data collection/processing**, image processing, and **hands-on experiments**.
Presented to the NSF REM Program my successful strategies for **effective mentoring** of research trainees.

Teaching Assistant — Worcester Polytechnic Institute
Robotics Engineering Department — **Aug 2019** – **May 2020**

Assisted in **delivery**, **labs**, and **grading** of robotics courses ('Actuation', 'Sensing', 'Introduction to Robotics').

Visiting Researcher — University of Ottawa
Mathematical Modeling & Physical Experiments — **Jun 2018** – **Aug 2018**

Modeled the vibrations of ribbed plates, and **experimentally** evaluated **analytical models** against **numerical models**.
My work resulted in **2 research publications** in reputable journals in the field of mechanical vibrations.

Research Assistant — Lebanese American University
Multiphysics Modeling & Numerical Simulation — **Nov 2017 – Mar 2019**

Demonstrated the viability of the 'ionic buoyancy engine' for underwater propulsion using a multiphysics **numerical model**. My work resulted in a research publication at a top-tier conference in the field of smart materials.

SKILLS

Software	C++, Python, Bash, Git, Linux, ROS2, Gazebo, MATLAB, Simulink, Mathematica, CMake
Technical	Dynamics, Control, Machine learning & AI, Motion planning, Computer vision, Algorithms
Research	Surveys/reviews, Mathematical modeling, Simulations, Physical experiments, Field work
Publishing	LaTeX, Technical writing, Photo-editing, Illustration/diagrams, Video-editing, MS Office
Social	Presentations, Communication (technical, general), Collaboration (cross-disciplinary)
Language	English (native fluency), French (professional fluency), Arabic (professional fluency)

PUBLICATIONS Preprints

[2023][1] Vision-based Framework for Fully-autonomous Oxy-fuel Metal Cutting [In Review, *IEEE/ASME TMECH*]
[2023][1] Cut Sequencing Algorithm for Safely Disassembling Large Structures [Accepted, *IEEE CDC*]
[2023][1] CNN-based Task State Estimation for Safer Automation of Oxy-fuel Metal Cutting [Accepted, *IEEE CASE*]
[2023][1] Vision-based Oxy-fuel Torch Control for Robotic Metal Cutting [Accepted, *IEEE/RSJ IROS*]
[2023][1] Feature-driven Next View Planning for Cutting Path Generation in Robotic Metal Scrap Recycling in ***IEEE T-ASE***
[2022][4] ZeroWaste Dataset: Towards Deformable Object Segmentation in Cluttered Scenes in ***IEEE CVPR***
[2021][1] Towards Robotic Metal Scrap Cutting: A Novel Workflow and Pipeline for Cutting Path Generation in ***IEEE CASE***
[2021][2] ECNNs: Ensemble Learning Methods for Improving Planar Grasp Quality Estimation in ***IEEE ICRA***
[2020][1] Comparing & Optimizing Analytical, Numerical & Experimental Vibration Models for [...] in ***IStructE/Elsevier Structures***
[2019][4] Simplified Setup for the Vibration Study of Plates with Simply-supported Boundary Conditions in ***Elsevier MethodsX***
[2019][1] Ionic Buoyancy Engines: Finite Element Modeling & Experimental Validation in ***SPIE Smart Structures***

[1–4] Superscripts indicate authorship position in co-authored publications.

PATENTS

Salvage Metal Cutting Robot · US 2022/03322003 A1 · Issued Oct 20, 2022
Autonomous Robotic Cutting System · 63/413,739 · Filed Oct 6, 2022
Feature-driven Next View Planning of 3-Dimensional Surfaces · 63/318,135 · Filed Mar 9, 2022

VOLUNTEERING

Reviewer — IEEE and Sage Publications
International Journals & Conferences — **Jun 2021 – Jul 2021**

Conducted scholarly peer review for reputed journals (IJRR, RA-L, T-ASE, TMECH) and conferences (IROS, CDC, CASE).

Faculty Candidate Reviewer — Worcester Polytechnic Institute
Robotics Engineering Department — **Feb 2023 – Mar 2023**

Attended **weekly seminars** of faculty candidates for **research and teaching** in the WPI Robotics Engineering Department. Submitted **written evaluations** of candidates' potential in research, teaching, supervision, funding, and inclusive work.

AWARDS

Dr. Glenn Yee Tuition Award, 2022
WPI Robotics Engineering Department

LAU Most Novel Club Award, 2018
Founded the Emerging Technologies Club

LAU Engineering Scholarship, 2015
Won a design and programming competition